GEM v

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1 Introduction

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General Emergence Mechanics

Persistence, Coherence Shells, and the Three Axes of Stable Existence Peter Baldwin 24th April 2026

Abstract

This paper introduces *General Emergence Mechanics* (GEM), a framework in which emergence is understood as the observable outcome of persistence within a coherence shell. The observable universe is modelled as a dynamical coherence shell anchored at the 255° vacuum minimum. Stable structures and patterns arise and persist only when three coupled axes—shell depth, stiffness, and viscosity/relaxation time—jointly support low-gradient coherence. Within this framework, time is not fundamental but emerges as the directional relaxation of the shell toward its preferred minimum. GEM provides a mechanical language for persistence, stability, degradation, and failure across physical, biological, chemical, computational, and engineered systems.

This paper develops the effective dynamical foundation of GEM, derives the three axes and their coupling, and shows how time itself can be understood as emergent relaxation within the shell. It then examines the framework's consistency with transient excitations and its initial computational expression in engineered systems.

2 Introduction – The Problem of Emergence

The question of emergence—why stable structures, patterns, and "things" persist in the universe rather than dissolving into noise—remains one of the deepest open problems in physics and philosophy. Existing approaches commonly treat emergence either as a brute fact, an epiphenomenon of complexity, or a higher-level description that does not reduce cleanly to underlying dynamics. In each case, the mechanism by which a configuration becomes persistent rather than transient is left unexplained or deferred to statistical arguments.

General Emergence Mechanics (GEM) proposes a different approach. It treats the observable universe as a dynamical coherence shell anchored at the 255° vacuum minimum. Within this shell, stable existence is not assumed but arises as the consequence of three coupled dynamical axes: shell depth, stiffness, and viscosity/relaxation time. Together these determine whether a configuration can persist long enough to count as a stable entity.

The central claim of GEM is mechanical rather than philosophical: *emergence is governed by persistence, and persistence is governed by the shell's dynamical properties.*

3 The Effective Dynamical Foundation of GEM

The foundation of GEM is an effective phase-field description derived from the underlying MT/GCF vacuum structure near the preferred minimum. At the effective level, the shell is described not only by a potential but by a free-energy landscape together with a dissipative relaxation law. This makes the theory explicitly dynamical and provides a natural route to temporal asymmetry.

3.1 Effective Shell Free Energy

We define the effective shell free energy as

$$F[\phi] = \int d^3x \left[\frac{1}{2}\nabla\phi^2 + V(\phi)\right], \tag{1}$$

where the potential is centered on the preferred vacuum phase

$$\phi_0 = 255^\circ = \frac{17\pi}{12}.$$

The effective shell potential is taken to be

$$V(\phi) = \Lambda^4\left[1 - \cos(\phi - \phi_0)\right] + \frac{\epsilon\Lambda^4}{2}\left[1 - \cos\big(24(\phi - \phi_0)\big)\right]. \tag{2}$$

The first term defines the primary basin around the 255° minimum. The second term is an exploratory orbifold-bias ansatz introducing a 24-fold sectorisation around the preferred phase. It should be treated as an effective symmetry scaffold rather than as a fully derived microscopic statement.

3.2 Rayleigh Dissipation and Deterministic Dynamics

To model irreversible relaxation, we introduce a Rayleigh dissipation functional

$$R[\dot{\phi}] = \frac{\gamma}{2}\int d^3x\,(\partial_t\phi)^2, \tag{3}$$

with $\gamma > 0$ the effective viscosity coefficient.

The deterministic dissipative dynamics are then given by the gradient-flow equation

$$\gamma \frac{\partial \phi}{\partial t} = -\frac{\delta F}{\delta \phi}, \tag{4}$$

or explicitly

$$\gamma \frac{\partial \phi}{\partial t} + \frac{\partial V}{\partial \phi} - \nabla^2 \phi = 0. \tag{5}$$

This is the simplest effective formalism in which the shell relaxes irreversibly toward the preferred vacuum phase.

3.3 Derivatives at the Minimum

The first derivative of the corrected potential is

$$V'(\phi) = \Lambda^4 \sin(\phi - \phi_0) + 12\epsilon\Lambda^4 \sin\bigl(24(\phi - \phi_0)\bigr). \tag{6}$$

At the preferred phase $\phi = \phi_0$,

$$V'(\phi_0) = 0. \tag{7}$$

The second derivative is

$$V''(\phi) = \Lambda^4 \cos(\phi - \phi_0) + 288\epsilon\Lambda^4 \cos\bigl(24(\phi - \phi_0)\bigr), \tag{8}$$

so at $\phi = \phi_0$ one finds

$$V''(\phi_0) = \Lambda^4 + 288\epsilon\Lambda^4 = \Lambda^4(1 + 288\epsilon). \tag{9}$$

This defines the effective local stiffness

$$\kappa_{\text{eff}} = V''(\phi_0) = \Lambda^4(1 + 288\epsilon). \tag{10}$$

3.4 Linear Relaxation Timescale

Linearising about the preferred vacuum phase,

$$\phi(x,t) = \phi_0 + \delta\phi(x,t), \qquad |\delta\phi| \ll 1,$$

the deterministic dissipative equation becomes

$$\gamma \frac{\partial(\delta\phi)}{\partial t} + \kappa_{\text{eff}}\, \delta\phi = 0. \tag{11}$$

Its solution is

$$\delta\phi(t) = \delta\phi(0) e^{-t/\tau}, \tag{12}$$

with characteristic recovery time

$$\tau = \frac{\gamma}{\kappa_{\text{eff}}} \approx \frac{\gamma}{\Lambda^4(1 + 288\epsilon)}. \tag{13}$$

This is the basic effective timescale that governs shell recovery and underpins GEM's interpretation of time as emergent relaxation.

4 Stability and Persistence in the Coherence Shell

Once the shell free energy and dissipative law are defined, the stability of the coherence shell and the conditions for persistence can be analysed by studying perturbations about the vacuum minimum.

4.1 Linearised Shell Dynamics

Let

$$\phi(x,t) = \phi_0 + \delta\phi(x,t), \qquad \phi_0 = 255^\circ,$$

with

$$|\delta\phi| \ll 1$$

in the linear regime.

To leading order, the effective shell dynamics are governed by

$$\mathcal{L}_{\text{lin}} = \frac{\gamma}{2}(\partial_t \delta\phi)^2 - \frac{\lambda}{2}(\nabla\delta\phi)^2 - \frac{\kappa}{2}(\delta\phi)^2. \tag{14}$$

The corresponding linearised equation of motion is

$$\gamma\,\partial_t^2\delta\phi - \lambda\,\nabla^2\delta\phi + \kappa\,\delta\phi = 0. \tag{15}$$

For Fourier modes

$$\delta\phi(x,t) \propto e^{i(k\cdot x - \omega t)},$$

this yields the dispersion relation

$$\gamma\omega^2 = \kappa + \lambda k^2. \tag{16}$$

The shell is therefore linearly stable provided

$$\kappa > 0, \qquad \lambda > 0, \qquad \gamma > 0,$$

since these conditions ensure $\omega^2 > 0$ for all wavevectors k.

4.2 Correlation Length and Spatial Persistence

The competition between the local restoring term and the gradient penalty defines a characteristic correlation length

$$\xi = \sqrt{\frac{\lambda}{\kappa}}. \tag{17}$$

This is the natural spatial persistence scale of the shell. Perturbations on scales much larger than ξ remain coherent more easily, while perturbations on scales much smaller than ξ are strongly suppressed. The corresponding critical wavelength is

$$L_c \sim 2\pi\xi = 2\pi\sqrt{\frac{\lambda}{\kappa}}. \tag{18}$$

4.3 Large-Perturbation Confinement

Beyond the linear regime, the quartic correction in the shell potential becomes important:

$$V(\phi) = \frac{\kappa}{2}(\phi - \phi_0)^2 + \frac{\alpha}{4}(\phi - \phi_0)^4. \tag{19}$$

The restoring force is

$$F_{\text{restore}} = -\frac{\partial V}{\partial \phi} = -\kappa \Delta\phi - \alpha(\Delta\phi)^3, \qquad \Delta\phi = \phi - \phi_0. \tag{20}$$

The quartic term preserves confinement of the shell basin so long as the perturbation remains within the effective bound

$$|\Delta\phi| \sqrt{\frac{\kappa}{\alpha}}. \tag{21}$$

4.4 Effective Relaxation and Persistence Threshold

In the full GEM picture, persistence requires more than linear stability: a configuration must remain coherent for longer than the shell's characteristic recovery time. At the effective level, this may be expressed as

$$T_{\text{persist}} > \tau_{\text{shell}}. \tag{22}$$

For the uniform mode in the deterministic overdamped regime,

$$\delta\phi(t) = \delta\phi(0) e^{-t/\tau_{\text{shell}}}, \qquad \tau_{\text{shell}} = \frac{\gamma}{\kappa_{\text{eff}}}. \tag{23}$$

More generally, the shell response may be decomposed into a superposition of decaying Fourier modes, each relaxing toward the basin around the preferred minimum.

If the persistence functional begins at some initial value P_0 and decays exponentially in the linear regime, then the coherence lifetime is approximately

$$T_{\text{persist}} \approx \tau_{\text{shell}} \ln\left(\frac{P_0}{P_{\min}}\right). \tag{24}$$

This makes the persistence criterion explicit: a configuration persists as a stable entity only when its coherence lifetime exceeds the shell recovery timescale. Configurations that fail this inequality remain transient.

4.5 Refined Persistence Functional

A clean prototype persistence functional is

$$P[\phi] = \Theta \left(\frac{D}{D_0}\right)^a \left(\frac{\kappa}{\kappa_0}\right)^b \left(\frac{\tau}{\tau_0}\right)^c \exp\left(-\frac{|\nabla\phi|^2}{\epsilon^2}\right), \tag{25}$$

where:

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- D is the effective depth (energy barrier to escape the shell),
- $\kappa = V''(\phi_0)$ is the local stiffness,
- $\tau = \gamma/\kappa$ is the relaxation time,
- $a, b, c > 0$ are dimensionless scaling exponents whose values remain to be determined,
- D_0, κ_0, τ_0 are reference scales,
- Θ is a domain-specific normalisation constant.

The exponential factor is a soft coherence penalty suppressing large spatial gradients. A configuration is persistent when

$$P[\phi] > P_{\min}, \tag{26}$$

and non-persistent when

$$P[\phi] < P_{\min}. \tag{27}$$

In its simplest form, the functional is taken to increase monotonically with the shell variables, though more detailed analysis may later reveal regime-dependent behaviour, especially in the dependence on τ.

4.6 Lyapunov / Monotonicity Result

Define again the effective shell free energy

$$F[\phi] = \int d^3x \left[\frac{1}{2}|\nabla\phi|^2 + V(\phi)\right]. \tag{28}$$

Along the deterministic dissipative dynamics

$$\gamma\frac{\partial\phi}{\partial t} = -\frac{\delta F}{\delta\phi}, \tag{29}$$

the time derivative is

$$\frac{dF}{dt} = -\gamma\int d^3x \left(\frac{\partial\phi}{\partial t}\right)^2 \leq 0. \tag{30}$$

Thus F is a Lyapunov functional for the deterministic shell dynamics: it is monotonically non-increasing and bounded below. This proves Lyapunov stability of the basin around the preferred phase ϕ_0 and shows that the shell possesses a preferred relaxation direction in free-energy space.

At the effective level, this provides the mathematical backbone for GEM's claim that temporal asymmetry emerges from shell relaxation. The arrow of time is not imposed externally, but appears as the preferred direction of dissipative descent in the shell free-energy landscape.

4.7 Formalised Metastability Taxonomy

Configurations in GEM may be classified by their persistence behaviour using both the persistence functional $P[\phi]$ and the local structure of the free energy $F[\phi]$:

[leftmargin=1.5em]

- **Stable:**
$$P[\phi] \gg P_{\min}$$
and ϕ sits in a deep local minimum of F with no lower nearby basin accessible under ordinary perturbations. Such configurations remain robustly occupied over long timescales.

- **Metastable:**
$$P[\phi] > P_{\min}$$
but ϕ lies in a local minimum separated from a lower basin by a finite barrier. Such configurations can persist for long periods, but may eventually tunnel, jump, or creep into the lower state.

- **Transient:**
$$P[\phi] \approx P_{\min}.$$
The configuration exists only briefly before relaxing away; it does not achieve robust occupation of the shell.

- **Non-persistent:**
$$P[\phi] < P_{\min}.$$
The configuration decays essentially immediately and behaves as a virtual-like fluctuation rather than a stable entity.

This taxonomy makes GEM operationally testable, since systems can be compared against the predicted signatures of stable, metastable, transient, and non-persistent behaviour.

5 Time as Emergent Relaxation

Within GEM, time is not treated as a fundamental backdrop in which the coherence shell exists. Rather, it emerges from the shell's own dynamics, with its temporal character set by the viscosity/relaxation axis.

A local perturbation displaces the vacuum phase from the 255° minimum,

$$\phi(x,t) = \phi_0 + \delta\phi(x,t), \qquad \phi_0 = 255^\circ.$$

Because the shell possesses finite viscosity, the perturbed phase does not return instantaneously to equilibrium. Instead, it relaxes toward the minimum over a

finite timescale. In the effective shell description, this characteristic recovery time is

$$\tau_{\text{shell}} = \frac{\gamma}{\kappa_{\text{eff}}}. \tag{31}$$

In the linear regime, the uniform mode decays exactly as

$$\delta\phi(x,t) = \delta\phi(x,0)e^{-t/\tau_{\text{shell}}} \qquad (uniformmode), \tag{32}$$

or more generally as a superposition of decaying Fourier modes. The decay is unidirectional toward the basin around ϕ_0.

The finite interval required for this return to equilibrium generates the distinction between "before" and "after." Temporal direction is therefore not imposed externally, but emerges from the shell's preferred relaxation toward lower free energy. In this sense, the arrow of time is the lived signature of finite-viscosity relaxation inside the coherence shell.

The **past** consists of those distortions that have already relaxed enough to leave a lasting record in the shell's configuration. The **present** is the instantaneous local posture of the shell—the current phase state. The **future** is the near-field of dynamically favoured but not yet completed relaxations: the paths the shell is gently pulling the system toward.

The limiting cases clarify the role of viscosity. In the limit

$$\gamma \to 0,$$

the shell would snap back instantaneously; there would be no temporal thickness, only reversible fluctuation. In the opposite limit

$$\gamma \to \infty,$$

the shell would remain permanently distorted and relaxation would cease; temporal unfolding would effectively freeze. Finite positive viscosity therefore produces the familiar unidirectional flow in which structures persist long enough to be experienced as stable.

More generally, the shell possesses a preferred relaxation direction because the effective free energy decreases monotonically:

$$\frac{dF}{dt} = -\gamma \int d^3x \left(\frac{\partial\phi}{\partial t}\right)^2 \leq 0. \tag{33}$$

Temporal asymmetry is thus the natural consequence of living inside a dynamical coherence shell with a preferred resting posture and a finite recovery time.

6 Virtual Particles and the Limits of Persistence

GEM provides a mechanical account of the distinction between transient excitations and stable entities. Within the coherence shell, the vacuum phase remains anchored near the 255° minimum. Local perturbations can briefly displace the

phase from this preferred posture, giving rise to short-lived excitations. However, not every excitation is capable of persistence. Only configurations that remain sufficiently phase-coherent with the shell's dynamical conditions can survive long enough to stabilise.

In this framework, virtual particles are interpreted as transient distortions of the vacuum phase. They arise when a local perturbation momentarily displaces the phase, but fail to satisfy the full persistence conditions imposed by the shell. Because the shell has finite depth, stiffness, and relaxation time, such excitations are driven back toward the vacuum before they can acquire stable identity.

The coherence shell therefore imposes a persistence threshold. Configurations that remain within the shell's basin and maintain sufficient coherence for longer than the characteristic relaxation timescale can stabilise as durable entities, including particles, atoms, and higher-order structures. Configurations that fall short of this threshold remain transient. The distinction between a virtual excitation and a stable particle is thus not one of kind, but of persistence.

At the effective level, the coherence lifetime is approximately

$$T_{\text{persist}} \approx \tau_{\text{shell}} \ln\left(\frac{P_0}{P_{\min}}\right), \tag{34}$$

where P_0 is the initial persistence amplitude and τ_{shell} is the characteristic shell relaxation time. The mechanical criterion then takes the explicit form

$$T_{\text{persist}} > \tau_{\text{shell}} \quad \Longleftrightarrow \quad stableentity, \tag{35}$$

$$T_{\text{persist}} < \tau_{\text{shell}} \quad \Longleftrightarrow \quad transientexcitation(virtualparticle). \tag{36}$$

This provides a clean mechanical criterion for existence. The shell acts as a natural filter: only sufficiently coherent, long-lived configurations survive to become observable "things." Short-lived distortions remain real at the level of local phase displacement, but they do not persist long enough to become stable components of the observable world.

7 Applications and Testable Predictions

GEM is intended not only as a theoretical framework but as a source of concrete, testable predictions across multiple domains. Its three coupled axes—depth, stiffness, and viscosity/relaxation time—together with the persistence threshold provide a common mechanical language for stability, resilience, degradation, and failure.

7.1 Materials and Viscoelastic Systems

In materials science, GEM maps naturally onto viscoelastic behaviour. Shell depth corresponds to the energy barrier that must be overcome for permanent deformation or fracture. Stiffness corresponds to the immediate elastic resistance of the material under load. Viscosity, or relaxation time, governs the rate at which stress relaxes, creep develops, or residual strain is retained.

This mapping suggests a clear set of predictions. Materials with high depth, high stiffness, and low viscosity should exhibit rapid recovery, low permanent deformation, and strong resistance under cyclic loading. Materials with high viscosity should instead show delayed recovery, long-term creep, hysteresis, or memory effects under repeated stress. Materials with low depth may appear locally stable under small loads but should fail abruptly once a critical threshold is crossed.

At the level of measurable quantities, one may write schematically

$$E_{\text{eff}} \sim stiffness, \qquad \tau_{\text{relax}} \sim viscosity, \qquad \Delta E_{\text{yield}} \sim depth,$$

so that persistence under cyclic stress is expected to depend on the joint condition

$$\Delta E_{\text{yield}} \gg k_B T_{\text{eff}}, \qquad \tau_{\text{relax}} \ll T_{\text{load}},$$

where T_{load} is the characteristic loading timescale. These predictions can be tested directly using standard rheological, creep, and fatigue measurements.

7.2 Biological and Chemical Homeostasis

In biological and chemical systems, GEM provides a mechanical language for resilience, regulation, and recovery. Cellular stress responses, immune regulation, metabolic control, and chemical buffering can all be interpreted as processes that maintain the local system near a stable persistence basin. Depth determines how far the system can be driven before irreversible breakdown or transition occurs. Stiffness governs the immediate buffering capacity against disturbance. Viscosity or relaxation time governs how quickly recovery occurs, and whether perturbations resolve cleanly or linger as chronic distortions.

GEM therefore predicts that biologically robust systems should display a favourable combination of sufficient depth, adequate stiffness, and moderate-to-low effective viscosity. Systems with high viscosity and reduced stiffness should be more prone to prolonged dysregulation, delayed recovery, or chronic pathological states. In chemical systems, similar distinctions should appear in buffering, equilibration, and recovery from perturbation.

At the schematic level, one may express resilience by a persistence inequality of the form

$$T_{\text{recovery}} < T_{\text{damage}}, \tag{37}$$

where T_{recovery} is the characteristic relaxation time of the system and T_{damage} is the timescale on which irreversible injury or transition occurs.

7.3 Computational Systems: Frankie as a Case Study

The Frankie v5–v7 architecture provides an initial computational case study for GEM-like dynamics. High-level external routing acts as a coarse coherence controller, while internal mechanisms such as phase-weighted attention, cache persistence shaping, and hidden-state evolution act as low-level persistence biases. Together they implement a selective persistence regime: costly operations

are preferentially invoked when conversational state becomes unstable, high-tension, or analytically demanding, while stable low-gradient cases are handled with reduced expenditure.

Across benchmark runs, these mechanisms produced measurable efficiency improvements relative to a monolithic baseline, including substantial reductions in token usage ($\approx 30--37\%$), latency ($\approx 40--47\%$), and overall work ($\approx 22--27\%$). The precise gains varied across versions and test sets, but the overall pattern remained stable: selective persistence and coherence-aware control improved computational efficiency without collapsing output quality.

A schematic computational analogue of the persistence condition may be written as

$$C_{\text{state}} > C_{\min} \quad \Longrightarrow \quad cheapstableprocessing,$$

$$C_{\text{state}} < C_{\min} \quad \Longrightarrow \quad costlycorrectiveprocessing,$$

where C_{state} denotes an effective coherence measure of the active conversational or representational state. Frankie therefore serves as an engineered proof-of-concept that GEM-style dynamics can be implemented, measured, and iteratively refined in practice.

In this way, GEM does not remain confined to speculative physical interpretation. It also offers a transferable mechanical framework for analysing and designing systems whose central problem is not mere computation, but persistent organised behaviour under stress.

8 Relation to MT/GCF and Broader Implications

General Emergence Mechanics (GEM) is not a separate theory from Modal Theory / the Global Coherence Framework (MT/GCF), but a direct development of it. MT/GCF provides the underlying vacuum structure: the 255° minimum, the orbifold geometry, the bias term, and the decoherence dynamics. GEM extracts the mechanical consequences of that structure for persistence, stability, and emergence.

Within this picture, the coherence shell is the natural basin formed around the 255° vacuum lock. The three axes—depth, stiffness, and viscosity/relaxation time—together with the persistence threshold, are not introduced as new fundamental postulates. Rather, they are the dynamical properties that arise once the vacuum is treated as a phase-stable basin with finite resistance to distortion and finite recovery time. In this sense, GEM is the operational mechanics of MT/GCF as applied to the question of why stable structures exist at all.

The relation may be expressed schematically as

$$MT/GCF \Longrightarrow vacuumstructure, \qquad vacuumstructure \Longrightarrow coherenceshell,$$

$$\text{coherence shell} \Longrightarrow persistenceconditions,$$

so that GEM functions as the effective mechanical layer through which MT/GCF is connected to the observable world of stable entities.

More formally, one may write

$$\mathcal{L}_{\mathrm{GEM}} = \mathcal{R}\big[\mathcal{L}_{\mathrm{MT/GCF}}\big]\,, \tag{38}$$

where $\mathcal{R}$ denotes the effective reduction map that expands the MT/GCF vacuum structure around the 255° minimum, coarse-grains temporal response into an effective viscosity γ, and projects the full field dynamics onto the three shell axes.

GEM therefore bridges the gap between MT/GCF's geometric and field-theoretic description and the observable world of persistent entities. It offers a common language of persistence across physics, chemistry, biology, information, computation, and engineered systems.

At the level of shared form rather than shared substrate, GEM proposes a transferable persistence structure of the schematic kind

$$stability \sim f(depth, stiffness, viscosity, persistencethreshold),$$

with different domains realising the same dynamical pattern in different physical, biological, chemical, or computational media. In this sense, GEM does not require that all systems share the same microscopic ontology; it requires only that they exhibit analogous persistence mechanics.

By providing a common language of persistence across domains, GEM opens the possibility of genuine cross-disciplinary transfer. Concepts, models, and experimental techniques developed in one field may illuminate others when translated into the language of shell depth, stiffness, viscosity, and persistence thresholds. In this framework, the coherence shell is not merely metaphorical. It is the central dynamical object through which emergence, stability, and failure can be described in unified mechanical terms.

9 Richer Structure and Natural Extensions

The three shell axes need not remain fixed global parameters. A richer GEM allows them to become local and history-dependent, turning the shell into a genuinely structured medium of persistence.

9.1 Depth, Stiffness, and Relaxation as Local Fields

In a richer formulation, the shell variables may be promoted to local fields,

$$D(x,t), \qquad S(x,t), \qquad \tau(x,t),$$

allowing the shell to become heterogeneous. Local weakening appears as a dip in D or S, regional recovery differences appear as spatial variation in τ, and long-lived coherence pockets or shell scars emerge as textured regions of enhanced persistence.

9.2 Shell Hysteresis and Memory

Once viscosity acquires path dependence, the shell can retain deformation history. This opens the door to hysteresis, cumulative stress, fatigue, delayed failure, chronic biological distortion, and computational drift. In this richer picture, persistence becomes explicitly history-dependent.

9.3 Defect Theory Inside the Shell

A dynamical basin naturally invites a defect language. Phase boundaries, local mislocks, and trapped distortions may become structured features rather than mere failures. Such defect-theoretic extensions could provide a natural bridge to materials science, chemistry, and biological organisation, though this remains a programme for further derivation rather than an established part of the present formal core.

9.4 Nested Persistence Domains

GEM also suggests the possibility of nested persistence domains across scales: vacuum, matter, informational, biological, and cognitive. These should presently be treated as exploratory extensions rather than as a settled hierarchy, but they point toward a broader mechanics in which persistence at one level may be embedded within persistence at another.

10 Conclusion – From Geometry to General Mechanics

GEM marks a shift from geometry to mechanics. What begins in MT/GCF as a description of the vacuum phase minimum becomes, in GEM, a dynamical framework for persistence and emergence. The coherence shell, with its three coupled axes of depth, stiffness, and viscosity/relaxation time, supplies the mechanical conditions under which configurations can persist long enough to count as stable entities. Time itself emerges as the directional relaxation of that shell.

The framework is both unifying and practical. It offers a single language that applies across physics, biology, chemistry, computation, and engineered systems. It explains why some excitations remain transient, why others stabilise, and how systems fail or recover under stress. The Frankie architecture provides an initial computational demonstration that GEM-like dynamics can be implemented and measured, yielding substantial efficiency gains without collapse of capability.

In GEM, emergence is not an extra principle layered on top of dynamics; it is what persistence looks like inside a coherence-supporting shell.

The framework does not claim to be a final theory. It is a mechanical starting point. Its central contribution is to replace vague appeals to complexity or higher-level description with a testable set of dynamical conditions: a persis-

tence threshold defined by the interplay of shell depth, stiffness, and relaxation time.

The next tasks are clear: derive the coupled equations more rigorously, quantify the persistence threshold, and test the framework's predictions in materials, biological systems, and engineered computation. If these steps hold, GEM may offer a transferable foundation for understanding not only how stable entities arise, but why persistence itself is the mechanical basis of emergence.

Bridge from Claim to Proof

The conceptual framework of GEM becomes a defensible theory only if its central claims are converted into explicit derivations. The immediate mathematical programme is therefore as follows.

A. Shell Stability

One must show that the 255° basin is dynamically stable. At minimum, this requires the local minimum conditions

$$V'(\phi_0) = 0, \qquad V''(\phi_0) > 0, \tag{39}$$

followed by the inclusion of gradient and coupling corrections. The goal is to show that small perturbations remain bounded or decay within the shell.

B. Persistence Threshold

The central criterion of GEM is that a configuration counts as stable only if its coherence lifetime exceeds the shell relaxation scale:

$$T_{\text{persist}} > \tau_{\text{shell}}. \tag{40}$$

More generally, one seeks a persistence functional

$$\mathcal{C}(\phi, \nabla\phi, \partial_t\phi; D, \kappa, \tau, \ldots) > \mathcal{C}_{\min}, \tag{41}$$

which separates transient fluctuation from persistent entity.

C. Temporal Asymmetry

Finite relaxation must be shown to generate directional evolution in state space. This requires comparing the zero-viscosity and high-viscosity limits and showing that finite intermediate viscosity yields ordered relaxation trajectories toward the basin minimum.

D. Transient vs Stable Excitations

The transient/stable distinction should be formalised by comparing excitation lifetime to shell relaxation time:

$$T_{\text{excitation}} < \tau_{\text{shell}} \Rightarrow \textit{transient}, \tag{42}$$

$$T_{\text{excitation}} > \tau_{\text{shell}} \Rightarrow \textit{stable}. \tag{43}$$

E. Cross-Domain Lawfulness

Finally, GEM's extension across physics, chemistry, biology, and computation should not be treated as a claim of identical microscopic substrate. Rather, the claim is that these domains may share a common *dynamical form*: basin depth, deformation resistance, recovery time, and persistence threshold.